Geometric Data Layouts and Resonance Stability via Golden Subdivision for Panmagic Square Chains

Abstract

We develop a mathematically coherent framework for a geometric data-layout scheme based on golden-ratio subdivision. The method is designed not as a fast matrix multiplication algorithm, but as a structural embedding method for arranging square data arrays, such as panmagic square chains, into elastic or toroidal media.

Given an $n \times n$ data manifold, we divide its coordinate directions using the golden ratio $\phi = \frac{1+\sqrt{5}}{2}$. Instead of splitting the domain equally into halves, we use an unequal decomposition with proportions $1/\phi$ and $1/\phi^2$. This produces a major block and a residual block whose sizes are maximally aperiodic in the sense of Diophantine approximation.

The main purpose of the construction is geometric stability. Equal rational subdivisions such as $1/2$ tend to create repeated alignments under deformation, which can amplify resonance in elastic layouts. Golden subdivision, by contrast, suppresses periodic coincidence because ϕ is the most poorly approximable irrational number. We formalize this as a stability hypothesis comparing the long-term deformation energy of golden-partitioned layouts against binary-partitioned layouts.

We also introduce a four-fold square-to-triangular packet construction. Under the assumption that each fold halves planar footprint and doubles layer multiplicity, four symmetric folds produce a packet with footprint ratio $1/16$ and 16 isomorphic layers. Finally, we clarify that cubic expansion $O(n^3)$ arises only when the folded structure supports $O(n)$ depth-scaled cross-layer interactions. Thus, the cubic term should be interpreted as a topological interaction cost, not as a direct consequence of having only 16 fixed layers.

1. Golden Block Partition

Let S_n be a square data domain with $n \times n$ discrete sites. Equivalently, let A and B be $n \times n$ matrices representing data fields on S_n.

Let

$$\phi = \frac{1+\sqrt{5}}{2}.$$

Then

$$\phi^2 = \phi + 1,$$

and therefore

$$\frac{1}{\phi} + \frac{1}{\phi^2} = 1.$$

We define a golden subdivision of the side length n by choosing integers m and r such that

$$n = m + r,$$

with

$$m \approx \frac{n}{\phi}, \qquad r \approx \frac{n}{\phi^2}.$$

For an actual finite array, one may choose a deterministic rounding rule, for example

$$m = \left\lfloor \frac{n}{\phi} + \frac{1}{2} \right\rfloor, \qquad r = n - m.$$

Asymptotically,

$$m \sim \frac{n}{\phi}, \qquad r \sim \frac{n}{\phi^2}.$$

Numerically,

$$\frac{1}{\phi} \approx 0.618, \qquad \frac{1}{\phi^2} \approx 0.382.$$

Thus the golden partition splits the square into four blocks:

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \qquad B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix},$$

where

- A_{11}, B_{11} have size $m \times m$;
- A_{12}, B_{12} have size $m \times r$;
- A_{21}, B_{21} have size $r \times m$;
- A_{22}, B_{22} have size $r \times r$.

The block A_{11} is the **major block**, while A_{22} is the **residual block**.

2. Block Multiplication and Computational Cost

The ordinary block product is

$$C = AB = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix},$$

where

$$C_{11} = A_{11}B_{11} + A_{12}B_{21},$$

$$C_{12} = A_{11}B_{12} + A_{12}B_{22},$$

$$C_{21} = A_{21}B_{11} + A_{22}B_{21},$$

and

$$C_{22} = A_{21}B_{12} + A_{22}B_{22}.$$

Using classical multiplication, the eight block products have costs

$$O(m^3), \quad O(m^2r), \quad O(m^2r), \quad O(mr^2), \quad O(m^2r), \quad O(mr^2), \quad O(mr^2), \quad O(r^3).$$

Adding these gives

$$m^3 + 3m^2r + 3mr^2 + r^3 = (m+r)^3 = n^3.$$

Therefore the golden block decomposition does **not** improve the asymptotic cost of classical matrix multiplication. Its computational cost remains

$$O(n^3).$$

This is important: the golden layout is not proposed as a faster algebraic matrix multiplication method. Instead, the partition is intended to improve **geometric embedding stability**.

3. Recursion and Topological Cost

If one recursively applies the golden partition only to the major block, then the dominant recursive size is approximately

$$\frac{n}{\phi}.$$

A recurrence of the form

$$T(n) = 5T\left(\frac{n}{\phi}\right) + O(n^3)$$

has exponent determined by

$$5\left(\frac{1}{\phi}\right)^p = 1.$$

Hence

$$p = \log_\phi 5.$$

Numerically,

$$p = \frac{\log 5}{\log \phi} \approx 3.345.$$

Therefore,

$$T(n) = O\left(n^{\log_\phi 5}\right) \approx O(n^{3.345})$$

if interpreted as a literal computational recurrence.

Thus, in this framework, such a recurrence should **not** be interpreted as a fast algorithm. Instead, it should be understood as a model of **topological expansion cost**: the extra interaction burden produced when a two-dimensional data manifold is folded into a three-dimensional layered medium.

We therefore distinguish two notions of cost:

1. **Algebraic computation cost**
   Classical block multiplication gives
   $$O(n^3).$$
2. **Geometric interaction cost**
   If $O(n^2)$ data sites are embedded into a medium with $O(n)$ depth-scaled interactions, the structural cost is
   $$O(n^2) \cdot O(n) = O(n^3).$$

The second quantity is the relevant cost in the physical-layout model.

4. Golden Subdivision and Anti-Resonance

4.1 Rational Splitting and Resonance

A binary split uses the rational ratio

$$\frac{1}{2}.$$

Repeated binary subdivision produces intervals of the form

$$\frac{k}{2^j},$$

where k and j are integers. Such points repeatedly align across scales. In a static computational grid, this is useful. But in a deforming elastic medium, repeated rational alignments may create coherent wave paths.

These coherent paths can amplify deformation energy, producing resonance. In a physical or simulated toroidal embedding, this may correspond to:

- repeated stress concentration,
- synchronized deformation modes,
- geometric tearing,
- local data-density collapse,
- or loss of panmagic adjacency structure.

Thus, while binary splitting is computationally convenient, it is geometrically resonant.

4.2 Golden Splitting and Irrational Aperiodicity

The golden ratio has continued fraction expansion

$$\phi = [1; 1, 1, 1, \dots].$$

This makes ϕ maximally resistant to rational approximation in the following sense: among irrational numbers, ϕ is the most poorly approximable by rationals. More precisely, rational approximations p/q to ϕ cannot converge too quickly.

This gives golden subdivision a useful anti-resonance property. A subdivision based on

$$\frac{1}{\phi}$$

does not repeatedly align with rational wave modes in the same way as binary splitting.

Therefore, golden subdivision introduces quasiperiodic spacing into the data layout. The resulting structure behaves more like a quasiperiodic tiling than a periodic grid.

5. Deformation Energy Model

Let L_α denote a data layout produced by repeated subdivision using ratio α, where

$$0 < \alpha < 1.$$

For binary splitting,

$$\alpha = \frac{1}{2}.$$

For golden splitting,

$$\alpha = \frac{1}{\phi}.$$

Let $E_\alpha(t)$ be the deformation energy of the layout L_α at time t under a specified elastic stress model.

A general form of the energy may be written as

$$E_\alpha(t) = \sum_{(i,j)\in\mathcal{E}} k_{ij} \left(\|x_i(t) - x_j(t)\| - \ell_{ij}^{(\alpha)} \right)^2,$$

where:

- $\mathcal{E}$ is the set of adjacency edges in the embedded data structure;
- $x_i(t)$ is the position of node i at time t;
- $\ell_{ij}^{(\alpha)}$ is the preferred rest length induced by subdivision ratio α;
- k_{ij} is the elastic stiffness associated with edge (i, j).

The long-term average deformation energy is

$$\overline{E}_\alpha = \limsup_{T\to\infty} \frac{1}{T} \int_0^T E_\alpha(t)\, dt.$$

The guiding stability hypothesis is:

$$\overline{E}_{1/\phi} < \overline{E}_{1/2}.$$

This is not a purely algebraic theorem. It is a geometric and physical hypothesis depending on the elastic model, boundary conditions, and forcing terms. However, it is mathematically motivated by the irrational anti-resonance of golden-ratio spacing.

6. Panmagic Square Chains

A panmagic square is a square array whose rows, columns, broken diagonals, and wrapped diagonals satisfy prescribed magic-sum constraints. These objects are naturally compatible with toroidal topology because their diagonal constraints wrap around the boundary.

Let P_n be an $n \times n$ panmagic square. The toroidal adjacency identifies opposite edges, so that positions are considered modulo n:

$$(i, j) \equiv (i + n, j) \equiv (i, j + n).$$

Thus the underlying discrete domain is

$$\mathbb{Z}_n \times \mathbb{Z}_n.$$

A panmagic square chain is a sequence

$$P_n^{(1)}, P_n^{(2)}, \ldots, P_n^{(q)}$$

embedded into a common toroidal or elastic medium, with compatibility maps between consecutive squares.

One may define chain links by maps

$$\Psi_k : P_n^{(k)} \to P_n^{(k+1)}.$$

The goal is to preserve panmagic constraints while minimizing deformation energy during embedding.

Golden subdivision is used to choose block boundaries and local packet structure so that the chain avoids repeated rational alignments.

7. Square-to-Triangular Folding and the $1/16$ Symmetry Law

7.1 Folding Assumptions

Let S be a square data manifold with area A_0 and n^2 discrete data sites.

Suppose there is a sequence of four folds

$$F = (f_1, f_2, f_3, f_4).$$

We assume each fold has two properties:

1. **Planar halving**
   Each fold halves the visible planar footprint:

$$\mathrm{Area}(f_i(S_{i-1})) = \frac{1}{2}\,\mathrm{Area}(S_{i-1}).$$

2. **Layer doubling**
   Each fold doubles the number of stacked layers:

$$L_i = 2L_{i-1}.$$

The initial surface has

$$A_0 = \mathrm{Area}(S), \qquad L_0 = 1.$$

7.2 Theorem: Four-Fold Symmetry Law

Theorem. Suppose a square data manifold S undergoes four folds, each of which halves planar footprint and doubles layer multiplicity. Then the final folded packet has planar footprint

$$A_4 = \frac{A_0}{16},$$

and consists of

$$L_4 = 16$$

isomorphic stacked layers.

Proof.

Since each fold halves area,

$$A_1 = \frac{A_0}{2}.$$

After two folds,

$$A_2 = \frac{A_0}{2^2}.$$

After three folds,

$$A_3 = \frac{A_0}{2^3}.$$

After four folds,

$$A_4 = \frac{A_0}{2^4} = \frac{A_0}{16}.$$

Similarly, since each fold doubles layer multiplicity,

$$L_1 = 2,$$

$$L_2 = 4,$$

$$L_3 = 8,$$

and

$$L_4 = 16.$$

Therefore the four-fold packet has footprint ratio

$$\frac{A_4}{A_0} = \frac{1}{16},$$

and layer count

$$L_4 = 16.$$

This proves the claim. $\square$

8. Square-to-Triangle Packet Construction

A square can be mapped into a right isosceles triangular footprint by a diagonal fold. Let the original square be

$$S = [0, 1] \times [0, 1].$$

The diagonal

$$x + y = 1$$

divides S into two congruent right isosceles triangles.

A diagonal fold identifies the two triangular halves, producing a triangular footprint of area

$$\frac{1}{2}.$$

Subsequent symmetric folds may be taken along medians or angle-bisecting axes of the triangular packet, provided each fold halves the footprint and doubles the layer count.

Thus, after four such folds, the resulting packet has:

- triangular or subtriangular planar footprint;
- footprint area $1/16$ of the original square;
- 16 stacked data layers;
- layerwise isomorphism induced by fold symmetry.

This packet is suitable as a local link in a toroidal panmagic chain because triangular simplicial packets can tile or wrap through a curved surface more flexibly than rigid square blocks.

9. Cubic Expansion as Depth Interaction

The four-fold law gives 16 layers, but a fixed number of layers alone does **not** imply cubic complexity.

Indeed, if an $n \times n$ data surface is folded into exactly 16 layers, the storage size remains

$$16O(n^2) = O(n^2).$$

Therefore, to obtain cubic scaling, one must introduce a depth-scaled interaction rule.

Let $N = n^2$ be the number of data sites. Suppose each site interacts not merely with a constant number of layers, but with $O(n)$ depth states, stress modes, or vertical coupling channels. Then the total interaction count becomes

$$O(n^2) \cdot O(n) = O(n^3).$$

This motivates the following definition.

Definition: Cubic Expansion Cost

The **cubic expansion cost** of a folded data manifold is the interaction complexity

$$C_{\text{exp}}(n)$$

arising when $O(n^2)$ planar data sites acquire $O(n)$ depth-scaled coupling channels after embedding into a three-dimensional medium.

Thus,

$$C_{\text{exp}}(n) = O(n^3).$$

In the present model, the 16 layers provide the local symmetric packet structure, while the $O(n)$ depth interactions arise globally from the panmagic chain, toroidal wrapping, or elastic deformation modes.

10. Golden Subdivision Inside the Folded Packet

The golden partition is applied before or during folding. The square is divided into major and residual regions with proportions

$$\frac{1}{\phi} \quad \text{and} \quad \frac{1}{\phi^2}.$$

Within the folded packet, these regions are distributed across the 16 layers. Because the partition boundaries are not rationally aligned with the binary fold structure, the final layout combines two different symmetries:

1. **Dyadic fold symmetry**
   The four folds create
   $$2^4 = 16$$
   symmetric layers.
2. **Golden subdivision asymmetry**
   The internal block boundaries follow the irrational ratio
   $$\frac{1}{\phi}.$$

The dyadic folding provides structural balance, while the golden subdivision prevents perfect resonance alignment between layers.

This is the central geometric idea:

$$\text{dyadic folding for symmetry} \quad + \quad \text{golden subdivision for anti-resonance.}$$

11. Stability Hypothesis for Panmagic Chains

Let $L_{1/2}$ be a panmagic square chain embedded using binary subdivision, and let $L_{1/\phi}$ be a chain embedded using golden subdivision.

Let their long-term deformation energies be

$$\overline{E}_{1/2}$$

and

$$\overline{E}_{1/\phi}.$$

We state the main stability conjecture:

Golden Stability Conjecture

For sufficiently large panmagic square chains embedded in an elastic toroidal medium with quasiperiodic forcing, the golden-subdivided layout has lower long-term deformation energy than the binary-subdivided layout:

$$\overline{E}_{1/\phi} < \overline{E}_{1/2}.$$

The justification is that binary subdivision produces rational alignment, while golden subdivision produces quasiperiodic spacing. Rational alignment supports coherent resonance; quasiperiodic spacing disrupts it.

12. Coherent Final Model

The complete model can now be summarized as follows.

1. Begin with an $n \times n$ panmagic square or matrix data field.
2. Partition each coordinate direction using

$$m \approx \frac{n}{\phi}, \qquad r \approx \frac{n}{\phi^2}.$$

3. This creates a major block, two rectangular minor blocks, and a residual block.
4. Classical block multiplication remains cubic:

$$O(n^3).$$

5. Four symmetric folds produce a layered triangular packet satisfying

$$A_4 = \frac{A_0}{16}, \qquad L_4 = 16.$$

6. The 16 layers provide local symmetry, while the golden subdivision prevents repeated rational alignment.
7. Cubic expansion appears only when the folded packet participates in $O(n)$ depth-scaled interaction modes:

$$O(n^2) \cdot O(n) = O(n^3).$$

8. The expected geometric benefit is reduced resonance energy:

$$\overline{E}_{1/\phi} < \overline{E}_{1/2}.$$

Final Formulation

Geometric Data Layouts and Resonance Stability via Golden Subdivision for Panmagic Square Chains is therefore best understood as a geometric embedding framework rather than a fast matrix multiplication algorithm.

Its central principle is:

$$\boxed{\text{Use golden-ratio subdivision to break resonance, and use four-fold symmetry to create stable layered packets.}}$$

The mathematically coherent version of the theory rests on four valid pillars:

- Golden partition:

$$\frac{1}{\phi} + \frac{1}{\phi^2} = 1.$$

- Four-fold footprint reduction:

$$\left(\frac{1}{2}\right)^4 = \frac{1}{16}.$$

- Four-fold layer multiplication:

$$2^4 = 16.$$

- Cubic interaction cost under depth scaling:

$$O(n^2) \cdot O(n) = O(n^3).$$

The resulting structure is a quasiperiodic, folded, toroidal data layout intended to stabilize panmagic square chains against elastic resonance.